

YAN LUO

Humanoid Robotics • Robot Learning • Embodied Foundation Models

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RESEARCH PROFILE

Ph.D. candidate in Mechanical Engineering at Shanghai Jiao Tong University, focusing on humanoid robotics and embodied robot learning. Research spans robust whole-body locomotion, language-conditioned reinforcement learning, VLA adaptation and RL post-training, structure-aware RGB-D perception, teleoperation, and real-world manipulation. Experienced in developing end-to-end systems from robot data collection and GPU-parallel simulation to real-time deployment on humanoid and dual-arm platforms. First-author publications include IEEE Robotics and Automation Letters and Science China Technological Sciences, with additional work under review. Reviewer for IEEE Robotics and Automation Letters.

Research Interests: Humanoid control and reinforcement learning; Embodied foundation models and VLA; Scalable robot data generation and policy refinement; Multimodal perception and manipulation.

EDUCATION

Shanghai Jiao Tong University (SJTU), Shanghai, China

Ph.D. Candidate in Mechanical Engineering

Sep. 2021–

Sep. 2026

Dissertation: Research on Semantic Perception, Dynamic Stability, and Policy Learning Methods for Humanoid Robots in Harvesting Tasks

Advisors: Prof. Han Ding (Member of the Chinese Academy of Sciences) and Prof. Zhenhua Xiong

Yanshan University (YSU), Hebei, China

B.Eng. in Automation; ranked 1st of 373 students

Sep. 2017–

Jun. 2021

SELECTED RESEARCH EXPERIENCE

MotionVL: Vision-Language Supervision for Humanoid Reinforcement Learning

2026

Humanoid robot learning, multimodal supervision, reinforcement learning

- Developed a closed-loop framework integrating task instructions, VLM-based motion diagnosis, LLM-generated reward functions, and PPO policy optimization.
- Evaluated straight-knee walking, running, continuous jumping, and single-foot balance across multiple reward candidates and random seeds; deployed straight-knee locomotion on a physical humanoid robot.
- Established a structured reward dictionary and numerical gating mechanism to improve reward reliability and reduce manual reward-engineering effort.

DCM-TVDW: Dynamic Stable Walking under Terrain-Induced Time-Varying Disturbances

2025

Humanoid dynamics, capturability analysis, robust locomotion control

- Proposed a DCM-based walking controller combining hybrid support–flight dynamics, N-step capturability, variable-height stabilization, and online coordination of center-of-mass height, step height, foothold, and gait phase.
- Validated dynamic recovery and locomotion stability on flat ground, moving platforms, rough terrain, and grass, with simulation and physical experiments.
- Developed a real-time convex optimization scheme for coordinated adjustment of center-of-mass height, step clearance, foothold, and gait phase.

TOVEP: Visual–Geometric Enhancement for VLM-Guided Robotic Picking

2026

Visual-geometric RGB-D perception, structure-guided VLM reasoning, task-oriented picking pose generation

- Proposed TOVEP, a task-oriented visual-geometric evidence enhancement framework that transforms noisy RGB-D observations into refined depth, structural-support, and confidence representations, explicitly grounding VLM reasoning in actionable geometry.
- Introduced structure-aware candidate generation and confidence-guided semantic inference to identify actionable grasp/cut attachment regions beyond conventional object-center localization, improving robustness to occlusion, reflective surfaces, missing depth, and visually ambiguous plant structures.
- Established a multi-crop RGB-D evaluation set and demonstrated consistent gains over RGB-only and raw-depth baselines without VLM fine-tuning, providing a plug-and-play geometric grounding interface for embodied manipulation.

Teleoperation and Vision-Language-Action Manipulation

2024

Generalist robot foundation models, scalable embodied data generation, reinforcement-learning-based policy refinement

- Developed a unified VLA training platform with an integrated UI, covering teleoperation-based data collection, dataset processing and inspection, training configuration, checkpoint management, policy deployment, robot reset, and closed-loop evaluation.
- Built a modular VLA training stack supporting flow-based policy adaptation, action chunking, scalable rollout collection, and reinforcement-learning post-training for long-horizon manipulation.
- Integrated RGB-D perception, VLM task reasoning, visual servo control, and ROS/SDK interfaces into a closed-loop perception–policy–execution pipeline for dual-arm and humanoid manipulation.

SELECTED PUBLICATIONS

(Candidate name shown in bold; publication status indicated where applicable)

1. **Yan Luo**, Zexi Bo, Ruonan Liu, Zhenhua Xiong, and Han Ding. "TOVEP: Visual–Geometric Evidence Enhancement for VLM-Guided Fruit Picking-Point Localization." Manuscript under review at *Computers and Electronics in Agriculture*.
2. **Yan Luo**, J. Wu, Z. Xiong, and H. Ding. "MotionVL: Vision-Language Supervision for Reinforcement Learning of Humanoid Motion." *IEEE Robotics and Automation Letters*, vol. 11, no. 5, pp. 5717–5724, 2026.
3. **Yan Luo**, C. Liu, J. Wu, Z. Xiong, and H. Ding. "DCM-Based Dynamic Stable Walking under Terrain-Induced Time-Varying Disturbances for Humanoid Robots." *Science China Technological Sciences*, vol. 68, no. 4, article 1420301, 2025.
4. F. Pang, Y. Chen, **Yan Luo**, Z. Lv, X. Sun, X. Xu, and M. Luo. "A Fast Obstacle Detection Algorithm Based on 3D LiDAR and Multiple Depth Cameras for Unmanned Ground Vehicles." *Drones*, vol. 8, no. 11, 2024.
5. **Yan Luo**, G. Chen, C. Liu, and Z. Xiong. "Image Foreground Segmentation Based on Small Data Set for Visual Servo Applications." Proceedings of the *IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, Seattle, USA, pp. 715–720, 2023.
6. G. Chen, **Yan Luo**, B. Huang, et al. "BP Neural Network–Fuzzy Control-Based Uncalibrated Visual Servoing Technology for Robots." *Science, Technology and Engineering*, 2023.
7. M. Zhang, **Yan Luo**, G. Chen, et al. "Design and Research of an Intelligent Maintenance Robot for a Nuclear-Power-Plant Steam Generator." *Science, Technology and Engineering*, vol. 23, no. 22, pp. 9559–9566, 2023.

RESEARCH LEADERSHIP AND SELECTED PROJECTS

- **Wheeled Humanoid Robot Development** — Led end-to-end platform development, spanning mechanical design, hardware integration, motion control, and manipulation; culminated in a piano-playing demonstration featured in national news media.
- **Humanoid Dynamic Balance Control** — Technical lead for projects supported by the Jiangsu Provincial Frontier Leading Technology Basic Research Program and the Jiangsu Industrial Transformation and Upgrading Program; led controller development, simulation, and physical validation.
- **Vision–Language–Action Robot Learning** — Led teleoperation data pipelines, VLA training and adaptation, RGB-D perception, visual servoing, and real-robot deployment on humanoid and dual-arm platforms.

PATENTS AND STANDARD

- Technical Requirements for Humanoid Robotics, T/CIET 648–2024, principal contributor, 2024.
- A Method for Remote Operation Control of a Semi-Humanoid Robot Based on VR Devices, authorized invention patent, second inventor, 2025.
- A Visual Servo Method Based on Multiple Image-Feature Fusion, authorized invention patent, second inventor, 2022.
- A Cross-Modal Alignment Vision-Language Navigation Method in Dynamic Occlusion Environments, published invention patent application, third inventor, 2025.

SELECTED HONORS AND AWARDS

- Outstanding Graduate, Shanghai Jiao Tong University (2026)
- First-Class Academic Scholarship, SJTU, 2022–2025 (four consecutive years)
- National Bronze Award, China "Internet+" Competition (2019)
- Jiangsu Province Young Talent Support Program (2025)
- National Scholarship (top 0.2%) (2020)
- First Prize, National Mathematical Modeling Competition (2019)

SELECTED METHODS AND ROBOTIC SYSTEMS

Humanoid Control and Learning: Dynamics and control formulation, DCM/capturability analysis, hybrid locomotion, GPU-parallel reinforcement learning, and sim-to-real deployment on physical humanoid robots.

Embodied Robot Learning: Multimodal VLA and language-supervised policy learning, flow/diffusion action policies, teleoperation-driven data pipelines, and reinforcement-learning post-training for long-horizon manipulation.

Perception-to-Action Systems: Structure-aware RGB-D grounding, uncertainty-aware semantic reasoning, visual servoing, and dual-arm coordination; Unitree G1 and custom robotic platforms; PyTorch, Isaac Lab/Sim, MuJoCo, ROS/SDK2/DDS, Python/C++, and Linux.

ACADEMIC SERVICE AND ACTIVITIES

- Reviewer for IEEE Robotics and Automation Letters.
- Scheduled to present MotionVL at the 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Pittsburgh, USA, Sep. 28–30, 2026.
- Invited technical presentation at the World Intelligent Manufacturing Conference on humanoid robot technologies.
- Presented research at the IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM), Seattle, USA.